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1. Array 1

Element	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

Target Number = 15

Linear Search: checks each element in order, one

10, 11, 12, 13, 14, (15)

Binary Search: checks middle number and checks if greater than or less than target

17, 13, (15)

2

2. $2 + 3 - (4 * 6) / 9$

Post-fix Conversion

Token	stack	Output
2		2
+	+	2
3	+	23
-	-	23+
(	(	23+
4	(	23+4
*	(*	23+4
6	(*	23+46
)	-	23+46*
/	(/	23+46*/
9	(/	23+46*9
	-	23+46*9/
		23+46*9/-

3.

a)	Element	"Adam"	"Bob"	"Charlie"	"Drew"
	Index	0	1	2	3
	"Eddie"	"Joseph"	"Killian"	"Xavier"	
	4	5	6	7	
	8	9	10	11	

Array Name: nameList

Helper Variable: count = 8

```

b) outputNames(nameList) {
    for (i = 0; i < count; i++) {
        output nameList[i]
    }
}

```

}

Results:

```

c) AdamLost(nameList) {

```

Bob

Charlie

Drew

Eddie

Joseph

Killian

Xavier

```

c) outFirstLast (nameList) {
    output nameList[0]
    output nameList[count-1]
}

```

Results:

Adam

Xavier

```

d) add Name (nameList, name) {
    i = 0
    while (i < count && nameList[i].charAt(0) <
           name.charAt(0)) {
        i++
    }
    for (index = count; index >= i; index--) {
        nameList[index] = nameList[index-1]
    }
    nameList[i] = name
    count++
}

```

Results:

name = "Fred"

Index	Element	Index	Element
0	Adam	7	Killian
1	Bob	8	Xavier
2	Charlie	9	
3	Drew	10	
4	Eddie	11	
5	Fred *		
6	Joseph		

4.

a) addVideo("Kubo", "Butler")

Title Array

Element	Kubo				Butler		
Index	0	1	2	3	4	5	6

addVideo("Spaceman", "Renck")

Director Array

Butler							
0	1	2	3	4	5	6	

* addVideo("Spaceman", "Renck")

Title Array

Kubo	Spaceman						
0	1	2	3	4	5	6	

Director Array

Butler	Renck						
0	1	2	3	4	5	6	

* addVideo("Missing Link", "Butler")

Title Array

Kubo	Spaceman	Missing Link					
0	1	2	3	4	5	6	

Director Array

Butler	Renck	Butler					
0	1	2	3	4	5	6	

* size()

Result: 3

* addVideo("Inception", "Nolan")

Title Array

Kubo	Spaceman	Missing Link	Inception			
0	1	2	3	4	5	6

Director Array

Butler	Renck	Butler	Nolan			
0	1	2	3	4	5	6

* addVideo("Shrek", "Adamson")

Title Array

Kubo	Spaceman	Missing Link	Inception	Shrek		
0	1	2	3	4	5	6

Director Array

Butler	Renck	Butler	Nolan	Adamson		
0	1	2	3	4	5	6

```
* remove("Butler")
```

Title Array

SpaceMan	Inception	Shrek				
0		2	3	4	5	6

Director Array

Reick	Nolan	Adamson				
0	1	2	3	4	5	6

```
* size()
```

result: 3

(b) WatchList will be implemented with two parallel lists: TitleArray and DirectorArray. There will be a helper variable "count" to keep track of the number of elements in both arrays. The size of the lists will be 7.

Title Array

Element	title1	title2					
Index	0	1	2	3	4	5	6

Director Array

Element	director 1	director 2					
Index	0	1	2	3	4	5	6
Count	= 2						

```
c) addVideo(title, director) {  
    TitleArray[count] = title  
    DirectorArray[count] = director  
    count++  
}
```

```
size() {  
    return count  
}
```

```
remove(director) {  
    for(i=0; i<count; i++) {  
        if (DirectorArray[i] == director) {  
            for (i2=i; i2<count; i2++) {  
                DirectorArray[i2] = DirectorArray[i2+1]  
                TitleArray[i2] = TitleArray[i2+1]  
            }  
            count--  
        }  
    }  
}
```

```
showList() {  
    for(i=0; i<count; i++) {  
        output DirectorArray[i]  
        output TitleArray[i]  
    }  
}
```